

FIT1051 Assessment One

Submission deadline: Friday 4th April 11:55pm MYT via Moodle

Weight: 12% of your overall unit mark

Late Penalty: 5% mark deduction per day

Instructions: Below are the coding tasks that you need to complete for Assessment 1. Your work and your submission should be independent. Please download the IntelliJ project folder below and unzip it. This will provide you with partial code in which to program your answers. Please complete each task in the appropriate section of the partial code, zip up the entire project folder, and submit to Moodle upon completion.

This assessment is worth 12% of the unit total. Submission of your code is worth 50% of this total, and your interview component (held later in Week 6) is worth 50%.

The programming portion of this assessment contains up to 50 marks and is comprised of the following components:

Code correctness is worth 50 marks:

- Task 1 is worth 25 marks
- Task 2 is worth 5 marks
- Task 3 is worth 20 marks

You must write good quality code that adheres to the FIT1051 [Coding Standards](#). You may be DEDUCTED up to 10 marks for not meeting these coding standards.

Academic Integrity: Please be reminded of the academic integrity standards that are expected of you at Monash, which were mentioned in Week 1. You should code alone and ask the unit staff for help if needed. Do not post your code in public forums or send your code to anyone. Do not copy/paste code from other sources and present it as your own — this includes use of generative AI tools. Breaching these academic integrity requirements can incur serious penalties.

Task 1: Design a Simple Class (25 marks)

(a) (10 marks) In the provided file WeatherData.java, create a new WeatherData class that represents the weather data for a single day. It will have fields representing the following attributes:

- the date, represented as an integer, in the form YYYYMMDD. For example, for the date 5th April 2025, the number will be 20250405;
- the minimum temperature for the day (may be decimal);
- the maximum temperature for the day (may be decimal);
- the rainfall amount in mm (only whole numbers);
- a simple free-text description of the day, for example “cloudy”, “sunny”, “foggy”.

Choose appropriate data types to represent each of these fields.

(b) (15 marks) Now, create the following methods for the WeatherData class:

- a default constructor, which initialises each field to a default value;
- a non-default constructor with five parameters to assign each attribute value;
- getter and setter method for each of the attributes;
- a toString method that returns the day’s information, formatted appropriately.

Task 2: Test Your Class (5 marks)

Once you have created this class, in the Assessment1 class under the task2() method, write code that tests your class:

- instantiates a WeatherData object, *day1*, using the default constructor;
- instantiates a WeatherData object, *day2*, for 17th February 2025, with a minimum temperature of 14, a maximum of 21, 2mm rainfall and description of “sunny”;
- uses the toString method to display each day.

Task 3: Loops, Data Input Validation and Array Processing (20 marks)

In the Assessment1 class under task3() method:

(a) (2 marks) Declare an array that will hold 7 temperatures;

(b) (3 marks) Write a loop to read from the user 7 different values for temperature, and store them in the 7 spots in the array;

(c) (5 marks) Adjust your loop so that each input value is validated before storing in the array. The temperature must be a value between 0 and 50 (inclusive), so if the user inputs a number outside that they should be told they have entered an invalid number and asked to re-enter it.

(d) (10 marks) Write code to process your array of temperatures to find the minimum and maximum temperature in the set. Display the results appropriately to the user.

Adherence to coding standards (DEDUCTION of up to 10 marks)

All code written should adhere to the guidelines set out in the FIT1051 [Coding Standards](#).

Submission Instructions: Please submit your IntelliJ project folder as a .zip file and submit to the Assessment 1 link on the Moodle Assessments page as shown below. If you are not sure how to zip your project, please refer to the [video](#) here. MAKE SURE YOU DOWNLOAD FROM MOODLE AFTER THAT TO CHECK IT IS THE RIGHT SUBMISSION!

Interview component: You will be asked to demonstrate your program at an interview in Week 6, following the code submission date. You will be contacted by your TA regarding the interview and if you need to book a time.

The interview will be 10 minutes in length, and consist of 6 questions. You may be asked to explain your code, your program designs, to modify your code, to discuss your coding decisions, or to explain the any of the coding concepts taught in Weeks 1-4 that this Assessment covers.

Interviews will take place in person during Applied class time, and also online via Zoom. You must have access to a stable internet connection and a working webcam, and your webcam must be switched on for the duration of the interview. Interviews will be recorded for marking integrity purposes, and recordings will be deleted at the end of semester.

It is your responsibility to make yourself available for an interview time. The interview is worth 50% of your assessment mark, and any student who does not attend an interview will receive a fail grade for the assignment. Your interview must take place before the end of Week 6.